THE ANSWERS

1. Compulsory question. These are multiple choice questions. Some questions can have multiple answers.

a) (i) False. (ii) False. (iii) True [3]

b) (iii) Forward bias, applied voltage is small [1]

c) (iii) the depletion width of the drain extending under the gate [1]

d) (i) Increasing the doping density in the substrate [1]

e) (ii) A region of the channel remains floating for all V_{GS} values [2]

f) (ii) II [2]

g) (iii) because of increased phonon scattering [2]

h) (ii) GAA-FETs are intended for the sub-7 nm node & (iii) GAA-FETs have better DIBL than finFETs [2]

i) (i) making the sheets wider, (iii) stacking more sheets vertically & (v) applying local tensile strain. [3]

j) (ii) Local strain: MOSFET is surrounded by a source and drain material with a different lattice constant; global strain: the channel is strained due to the substrate [2]

k) (ii) a change in the effective mass of the charge carrier in the channel, (iii) a change of the mobility of the charge carrier in the channel & (iv) a change in the shape of the energy band diagram of the strained layer. [3]

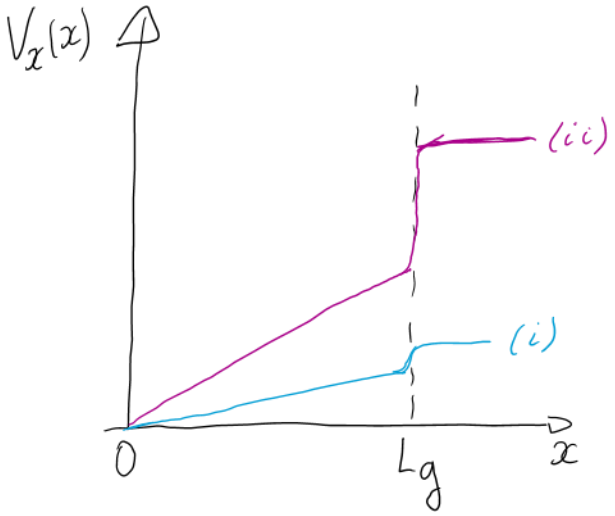
l) (ii) by right clicking on the “no tools” field, selecting “add” [1]

m) (ii) isotropically [1]

n) (iii) Force the calculation at the initial voltage on the electrode [1]

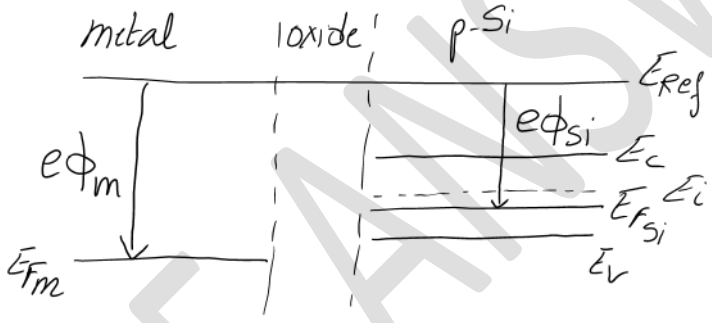
THE ANSWERS

2. a)

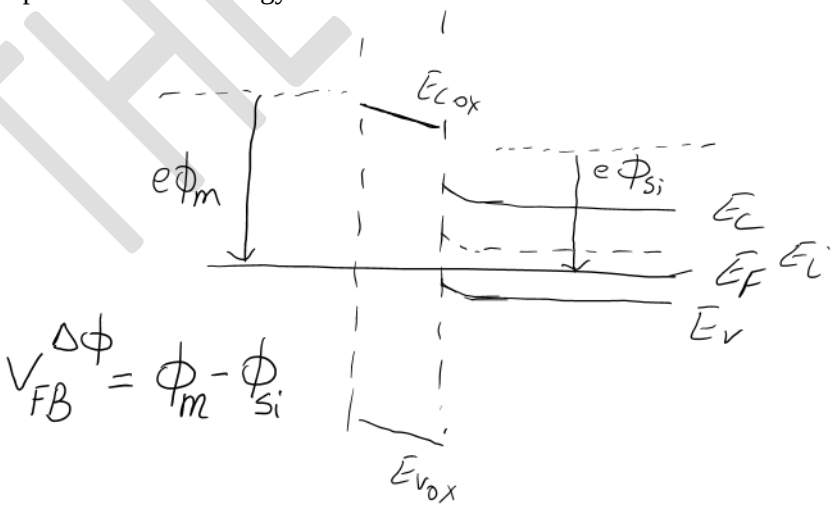


i) [2] & ii) [2]. Important is the difference in slope between 0 and Lg and the big sudden step at Lg that should be much bigger for (ii) than (i).

b) Start with taking the workfunction difference into account but not the charge in the oxide. Start with the plot before contact.



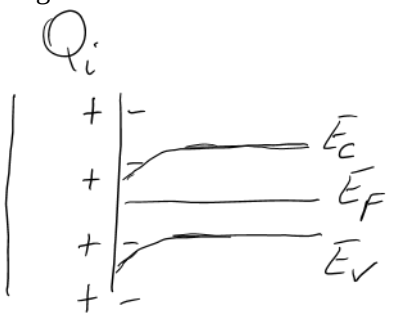
Upon contact the energy bands become:



[2]

THE ANSWERS

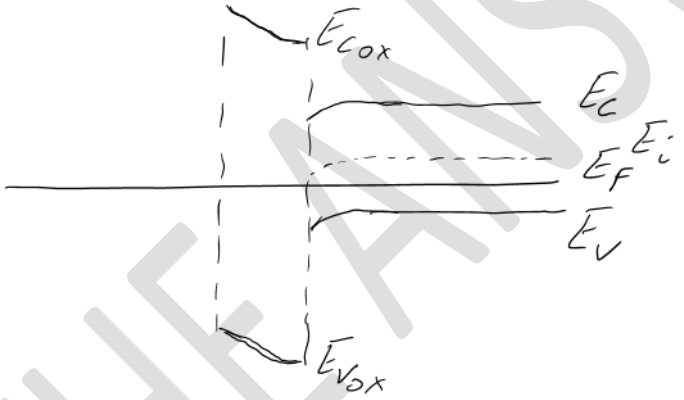
The bands in the semiconductor and oxide must band in the way drawn. In order to create flat band a positive voltage: $V_{FB} = \phi_m - \phi_{si}$ ($\Delta\phi$) must be applied.
Now take only the impact of the positive charge Qit into account when drawing the band diagram:



$$V_{FB}^{Q_i} = -\frac{Q_i}{C_{ox}}$$

[2]

A positive charge in the oxide must be compensate by a negative charge in the semiconductor. This means that the bend in the semiconductor must bend down. Now the voltage that needs to be applied for flat band is negative: $V_{FB}^{Q_i} = -\frac{Q_i}{C_{ox}}$
The combination of the two effect depends on the assumption that the students make of which V_{FB} is largest in positive value. Under the assumption that Q_{it} bends the bands more than $\Delta\phi$ then the final energy band diagram would be:



[4]

In principle the bending should be smaller than in the previous figure. In principle some non-linearity should occur in Ec and Ev at the end of the oxide near the Si interface due to the presence of charge. It this isn't taken into account, no marks will be subtracted.

c)

i) enhancement [2]

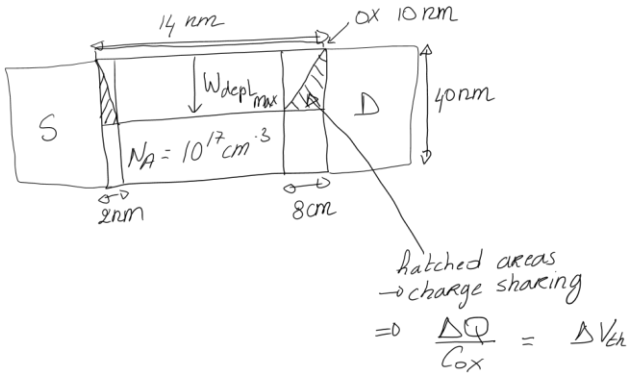
ii) the region where the log(I_{DS}) is linear [2]

iii) electron diffusion [2]

e)

THE ANSWERS

Start by plotting the structure and realising that this question is one of charge sharing. The problem is simplified because all geometries are rectangular. It is an nMOS thus the substrate is p-type.



[2]

The maximum depletion region extending from the gate must be extracted. Referring to the equations available in the formulae sheet and the constants in the constants list.

$\phi_F = V_T ln\left(\frac{N_A}{n_i}\right)$ the Fermi potential in the substrate needed to find the condition for threshold

$$W_{depl\,max} = \sqrt{\frac{2\varepsilon(V_{bi}-V)}{e}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)}$$

The maximum depletion width is reached at threshold -V = 2 ϕF

V_{bi} in MOS can be assumed to be 0 V

1-sided junction thus $1/N_D$ goes to zero.

$$W_{depl\,max} = \sqrt{\frac{4\varepsilon\phi_F}{eN_A}}$$ [2]

$\Delta V_{th} = \frac{\Delta Q_{depl}}{C_{ox}}$ & $C_{ox} = \frac{\varepsilon}{t_{ox}}$

V_T = 0.026 V

n_i = 1.45 10^{10} cm^{-3}

ε_0 = 8.85 10^{-12} F/m = 8.85 10^{-14} F/cm (best to use cm as units)

ε_{ox} = 4 & ε_{ox} = 12 & e = 1.6 10^{-19} C

$C_{ox} = \frac{\varepsilon}{t_{ox}} = \frac{4\times 8.85\ 10-14\ \mathrm{F/cm}}{10\ 10^{-7}\ cm} = 3.54\ 10^{-7} F/cm^2$

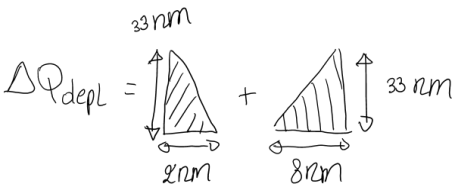
$C_{ox} \times W \times L_g$ = 14 $10^{-7} cm \times 20\ 10^{-7} cm \times 3.54\ 10^{-7} F/cm^2$ = 9.91 10^{-19} F

$\phi_F = V_T ln\left(\frac{N_A}{n_i}\right) = 0.026\ V \ln\left(\frac{10^{17}}{1.45\ 10^{10}}\right) = 0.409\ V$

$$W_{depl\,max} = \sqrt{\frac{4\varepsilon\phi_F}{eN_A}} = \sqrt{\frac{4\times 12\times 8.85\ 10^{-14}\frac{\mathrm{F}}{\mathrm{cm}}\times 0.409V}{1.6\ 10^{-19}C\times 10^{17}cm^{-3}}} = 3.3\ 10^{-6} cm = 33\ nm$$ [1]

Thus, the regions where charge sharing takes place is, taking a 50/50 approach of sharing between the gate and source & drain into account:

THE ANSWERS



$\Delta Q_{depl} = -eN_A t_j(2+8)W = -1.6\ 10^{-19}C \times 10^{17}cm^{-3} \times \frac{2+8}{2} 10^{-7}cm \times 33\ 10^{-7}cm \times 10 \times 40\ 10^{-7}cm = -5.29\ 10^{-20}C$

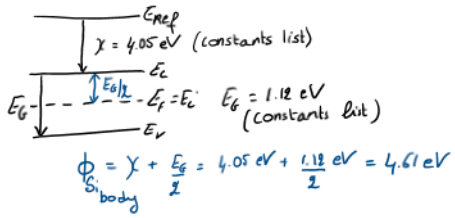
$\Delta V_{th} = \frac{\Delta Q_{depl}}{C_{ox} \times A} = \frac{-5.29\ 10^{-20}C}{9.91\ 10^{-19}F}$ = -53 mV. [2]

THE ANSWERS

2. a)

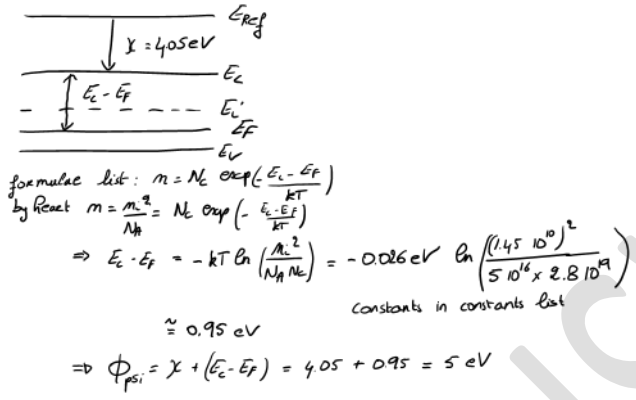
To ensure all relative variations of the potential energy are correct, it is best to calculate some parameters. The workfunction of the Si body and the Si substrate will need to be derived. Parameters can be found in the constants list:

Workfunction of the Si body:



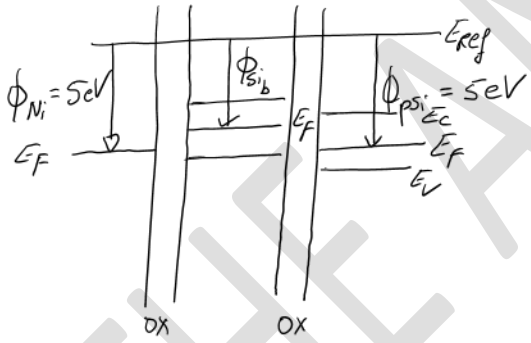
[1]

Workfunction of the Si substrate



[1]

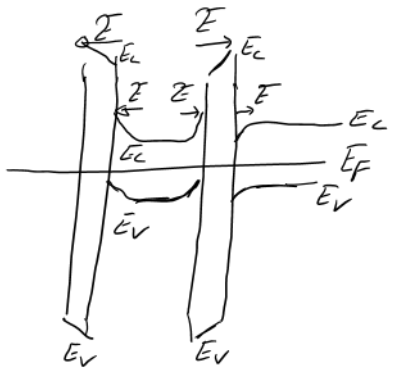
Plotting the energy band diagram before contact:



note that the workfunction of pSi and of Ni are the same under the given conditions.

Upon contact:

THE ANSWERS



Note E_c & E_v flat in middle of the channel because no volume inversion

[1] for constant EF
[2] for upwards bending of the same amplitude in the Si body at both ends
[1] for the correct slope in oxides
[2] for the correct pSi band diagram
[2] for the correct electric fields

b)

i) Yes. This is a pMOS (p-type doping in the ohmic contacts). The density of holes near the oxides is larger than the density of electrons in the middle of the Si body (cfr definition of Vth). [4]

ii) Depletion means the intrinsic carrier density occurs where no inversion occurs, thus FD. [2]

iii) Compressive, local strain. [2]

iv) Lattice constant Si: 0.543 nm [2]
Lattice constant of Ge: 0.565 nm
Lattice constant SiGe with 30% Ge: 0.543×0.7+0.565×0.3
Strain = $\varepsilon = \frac{a_{SiGe} - a_{Si}}{a_{Si}} = \frac{0.543 \times 0.7 + 0.565 \times 0.3 - 0.543}{0.543} = 0.012$ or 12% strain

v) It increases the hole mobility in the channel [2]

c) Flat band means that there should be no band bending. In this case the contact potential difference needs to be applied to the common gate: $V_{GS} = V_{FB}$ = (5 V – 4.61 V) = 0.39V. The sign of the voltage should be positive. [3]

THE ANSWERS

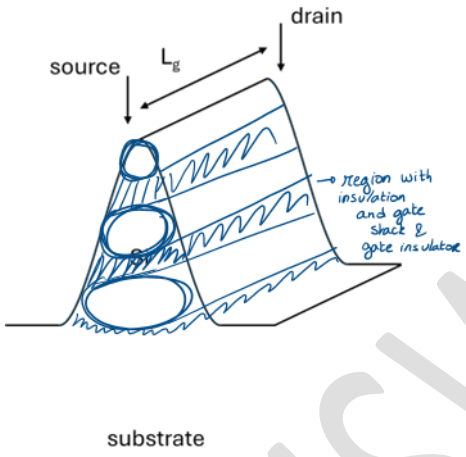
4.

a) a tapered fin is normally one that is defined on bulk due to the STI etch leaving a tapered rather than a rectangular fin. [4]

b) two [2]

c)

i) [4]



_____ of importance is that some indication is given of the insulation between the different nanowires and that the nanowires cannot all have the same size because of the taper. The shape can be round or square but they should not look like nanosheets.

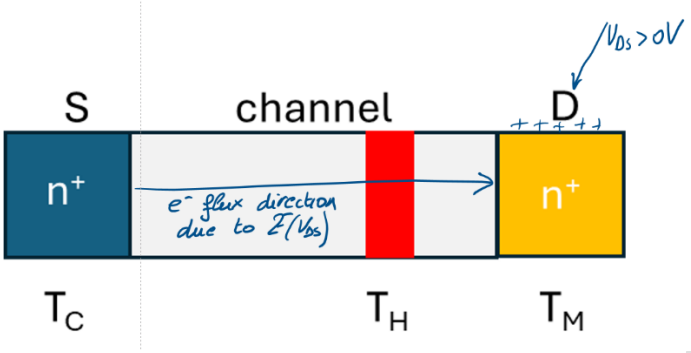
ii) It is expected that the density of carriers will be largest in the smallest nanowire because its V_{th} will be smaller. Unless of course there is volume inversion in this NW. [2]

iii) The NW with the smallest size will have the highest resistance but also the lowest thermal conductivity. Thus there self-heating will be highest and the removal of this heat will be least efficient. However it is near the top and thus the path for cooling is short. When looking at the bottom NW it has the largest diameter thus the highest thermal conductivity and also it sits near the bulk that offers a good way of cooling. The middle wire is surrounded by insulators with long cooling path length a relatively low thermal conductivity this the probably that it is hottest is highest. [4]

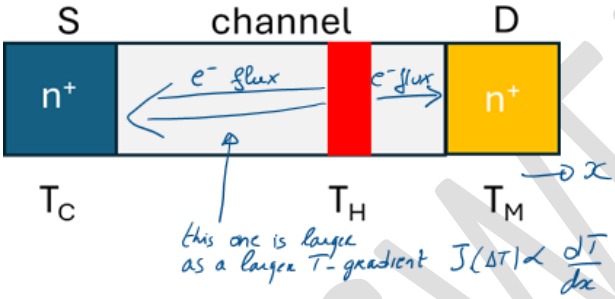
d)

i) [2]

THE ANSWERS



ii) [4]



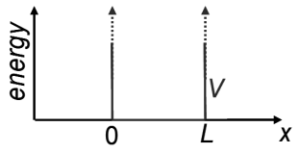
iii) decreasing [2]

THE ANSWERS

5.

(a) Potential energy given by:

$V(x) = 0, \ 0 \le x \le L$
$V(x) = \infty, \ x < 0, x > L$



[2]

Particle cannot exist ($\psi = 0$) where $V(x) = \infty$, $\Rightarrow$ particle trapped in the well

Within the well, space-dependent Schrödinger equation $\Rightarrow$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} - E\psi(x) = 0$$

General solution is of the form:

$$\psi = A\sin kx + B\cos kx \quad \text{where} \quad k = \sqrt{\frac{2mE}{\hbar^2}}$$

[4]

Boundary conditions:

Substituting $\psi = 0$ at $x = 0$ in the general solution $\Rightarrow$

$B = 0 \Rightarrow \psi = A\sin kx$

Substituting $\psi = 0$ at $x = L$ in this $\Rightarrow$

$$\psi = A\sin kL = 0 \quad \Rightarrow \quad kL = n\pi \quad \text{where } n = 0, 1, 2....$$

This gives the particle energy levels:

$$E_n = \frac{\hbar^2 k^2}{2m} = \frac{n^2\pi^2\hbar^2}{2mL^2}$$

The energies are quantised

[4]

Total [10]

(b) To find the full solution for ψ, we normalise the wave-function:

$$\int_{-\infty}^{+\infty} \psi^2(x)dx = 1 \quad \Rightarrow \quad \int_0^l A^2 \sin^2\left(\frac{n\pi x}{L}\right)dx = 1$$

$$\Rightarrow A^2 \int_0^l \frac{1}{2}\left(1 - \cos\frac{2n\pi x}{L}\right)dx = 1 \quad \Rightarrow A = \sqrt{\frac{2}{L}}$$

$$\Rightarrow \psi(x) = \sqrt{\frac{2}{L}}\sin\frac{n\pi x}{L}$$

[6]

Probability density functions $\psi_n\psi_n^*$ are then given by:

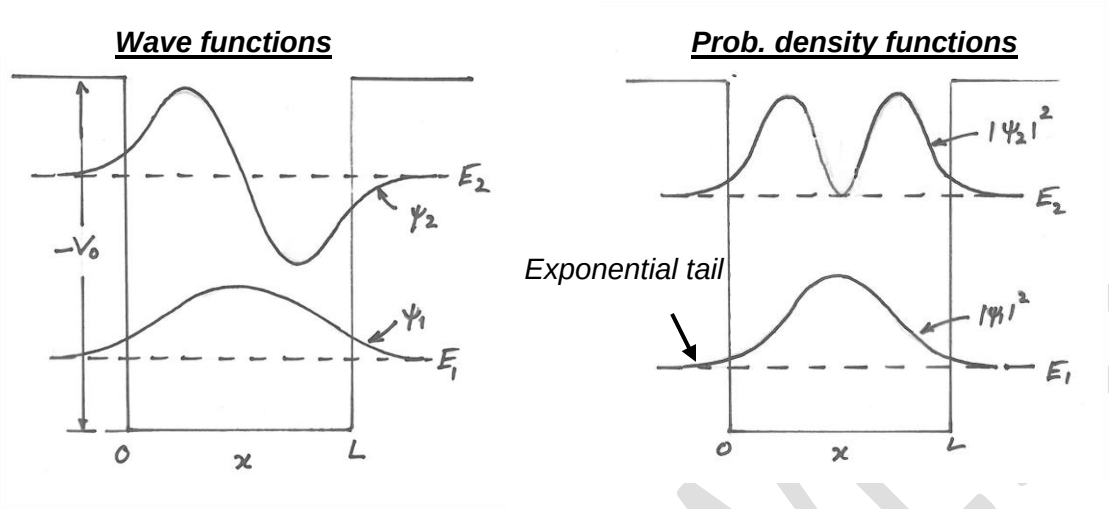
$$\psi_n\psi_n^* = \frac{2}{L}\sin^2\frac{n\pi x}{L}$$

[2]

THE ANSWERS

Total [8]

(c) Sketches of the wave functions, and the probability density functions, in the finite potential well for n = 1 and 2 are as follows:



3 marks for each figure

The peak value of the probability density function for n = 1 in the finite potential well is less than that in the infinite potential well. This is because there is a non-zero, decaying probability that the particle exists outside the well.

1 mark for similar argument

Total [7]

THE ANSWERS